Ben Sedran

Bedford MA, 01730
bws72@cornell.edu linkedin.com/in/Ben-Sedran Engineering Portfolio

Education

Cornell University **Aug. 2024 – May 2028**
Bachelor of Science in Mechanical Engineering *Ithaca, NY*
Minor in Computer Science **GPA: 3.7**

Relevant Coursework

- Dynamics and System Dynamics
- Statics and Mechanics of Materials
- Mechanical Design
- Thermodynamics and Fluid Mechanics
- Computer Systems Organization
- Functional Programming and Data Structures
- Object Oriented Programming and Data Structures
- The Mathematics of Computing

Research Experience

Santomartino Space-Microbiology Research Group | *Research Assistant - Ithaca NY* **May 2026 - Aug. 2026**

- Built 8 random positioning machines (RPMs) to simulate microgravity for microorganisms in future space plastic degradation experiments for $\sim$ \$100 each using 3D printing, Nema 17 steppers, TMC2209 drivers, and an Arduino.
- Engineered a modular interface supporting 7 container types and 6-24 samples per run, built to extend to future media.
- Measured microgravity effect by developing 8 compact, battery-powered g-vector loggers that log orientation and acceleration data to a microSD.
- Documented in a GitHub repo detailed build, maintenance, and use processes for the RPMs and g-vector loggers.

Project Team Experience

Cornell Autonomous Drone | *Mechanical Sub team* **Sept. 2026 - Present**

- Designing and fabricating an octocopter drone from carbon fiber and PLA with a high thrust to weight ratio.
- Researching and designing a drone with firefighting capabilities to fight dangerous and difficult to reach structure fires.

Other Work Experience

Pool Supervisor and Swim Instructor | *Billerica Boys and Girls Club, Billerica MA* **July 2021 - Aug. 2025**

- Taught group and individual swim lessons year round to children of all skill levels from ages 3 to 12.
- Was promoted to supervisor in 2024, and maintained chemical concentrations in the pool and administered swim tests.
- Trained new instructors to run group lessons efficiently and effectively teach while maintaining the safety of all students.

Lifeguard and Swim Instructor | *Spring Brook Park, Bedford MA* **June 2022 - Aug. 2025**

- Taught basic to intermediate swimming skills to large groups of town summer campers and lifegaurded.
- Responsible for several saves of struggling swimmers, and initiated and led missing child search protocols.

Projects

Digital Alarm Clock | *Arduino circuitry, soldering, C++, 3D printing* **Aug. 2026**

- Wrote C++ firmware for an RP2040 MCU that keeps time with a DS3231 RTC and drives a 7-segment display over I^2C.
- Wired a custom control panel with rotary encoders and switches to adjust clock and alarm time.
- Encased the circuitry and display in a custom plywood box, and 3D printed apparatus to keep it all in place.

Spotted Lanternfly Trap | *Laser cutting, 3D printing, Autodesk Fusion* **Feb. - May 2026**

- As a course project, engineered a mechanical trap around grape industry client needs to combat vineyard infestation.
- Created adjustable trap mechanisms for tuning during testing with actual lanternflies to maximize trap effectiveness.

Technical Skills

Programming Languages: Java, C/C++, OCaml, RISC-V Assembly
Tools and Technologies: AutoDesk Fusion, Onshape, MATLAB, Linux, VS Code, Git, Arduino IDE
Other Skills: 3D printing, Soldering, Machining, Woodworking

Extracurriculars / Interests

Cornell Club Water Polo **Aug. 2025 - Present**

- Member of the Cornell Club Water Polo Team, competing for the NY and National Club Water Polo championships.

Amateur Chess Player **March 2021 - Present**

- Played online consistently for five years with a 1900+ rapid rating on chess.com.